



Experiment 01: Agile Software Development Lifecycle using Jira

Experiment No.	01
Mapped CO	CO2
Aim	To study and implement the Agile Software Development Lifecycle using Jira by creating and managing a Scrum project, including product backlog, user stories, sprint planning, task assignment, and sprint tracking.
Tools / Software	Git (2.30); A GitHub account; A code editor / terminal

Theory

Agile is an iterative software development methodology that focuses on delivering software in small, incremental releases while continuously incorporating customer feedback. It promotes collaboration, flexibility, and rapid delivery of high-quality software.

Scrum is one of the most widely used Agile frameworks. It organizes work into fixed-duration iterations called **Sprints**, during which the development team completes selected tasks from the **Product Backlog**. Scrum defines roles such as **Product Owner**, **Scrum Master**, and **Development Team**, and includes events like Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective.

Jira, developed by Atlassian, is a popular Agile project management tool that enables teams to create projects, manage user stories, plan sprints, assign work, track progress using Scrum boards, and generate reports such as Burndown Charts and Sprint Reports.

Procedure

- ☐ Create a Jira Cloud account and log in.
- ☐ Create a new Scrum Software Project.
- ☐ Enter the project name and create the project.
- ☐ Create Epics for different functional modules.
- ☐ Create User Stories using the format **As a..., I want..., So that...**
- ☐ Add all stories to the Product Backlog.
- ☐ Create Sprint 1.
- ☐ Move selected User Stories into the Sprint.



- ☐ Start the Sprint by specifying its duration and Sprint Goal.
- ☐ Create Tasks/Subtasks for each User Story.
- ☐ Assign tasks to team members.
- ☐ Update task status from **To Do** → **In Progress** → **Done**.
- ☐ View Scrum Board to monitor project progress.
- ☐ Generate Burndown Chart and Sprint Report.

Observation / Experiment Output

Instructions to students: Paste screenshots or copied terminal outputs in the boxes below after completing the Git commands. Keep screenshots clear and readable.

Name	Roll No.	Batch	Date
Vighnesh Pingale	48	B2	10/07/26

Sample Code / Commands

Sample Activity

Project Name

Online Food Ordering System

Example User Story

Story: User Registration

As a new customer,

I want to register on the application,

So that I can order food online.

Example Sprint

Sprint Name : Sprint 1

Duration : 2 Weeks

Sprint Goal :

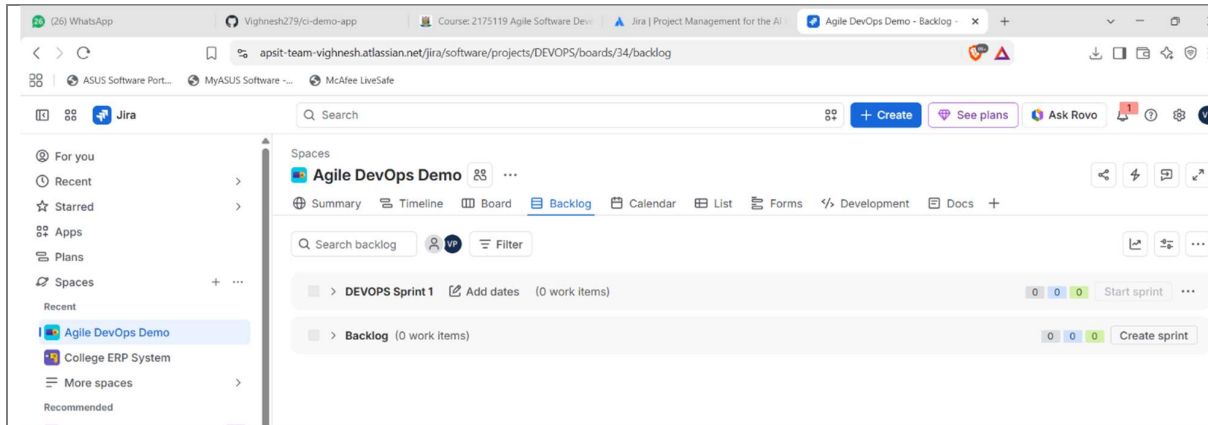
Complete User Authentication Module



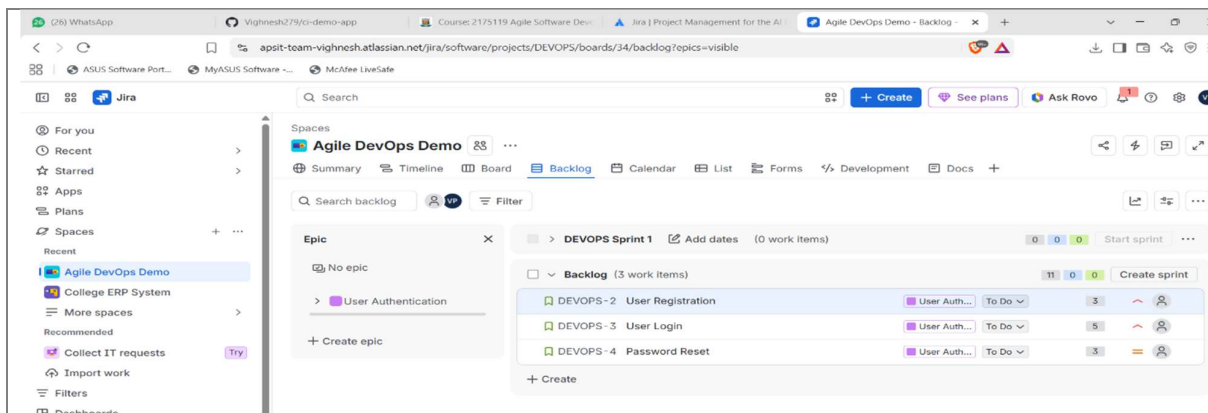
PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science



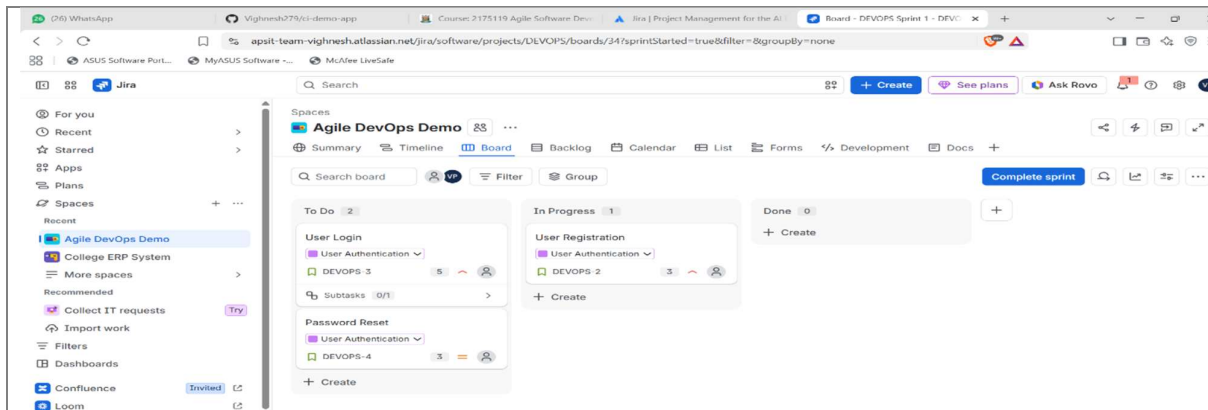
1. Jira Dashboard



2. Product Backlog



3. Sprint Planning





4. Scrum Board

The screenshot displays the Jira interface for a Scrum project named 'Agile DevOps Demo'. The main view is the Scrum Board, which shows a backlog of tasks categorized into 'To Do', 'In Progress', and 'Done'. The 'Done' column contains tasks such as 'User Registration', 'User Login', and 'Password Reset', each with a 'DEVOPS-2' label. To the right, the 'Comparing changes' section shows a pull request for 'DEVOPS-3 Add Jira GitHub integration P1'. Below this, the 'Burnup report' and 'Velocity report' are visible. The Burnup report shows a line graph of story points completed over time, with a table of events below it. The Velocity report shows a bar chart of story points completed per sprint, with a table of events below it.

Date	Event	Work Item	Completed	Scope
Fri, Sep 25, 2026, 4:08pm	Sprint started	DEVOPS-2 User Registration DEVOPS-3 User Login DEVOPS-4 Password Reset	0	18
Fri, Sep 25, 2026, 7:04pm	Work item completed	DEVOPS-2 User Registration	10	18
Fri, Sep 25, 2026, 7:04pm	Work item completed	DEVOPS-3 User Login	10	18
Fri, Sep 25, 2026, 7:04pm	Work item completed	DEVOPS-4 Password Reset	10	18
Fri, Sep 25, 2026, 7:04pm	Sprint completed	DEVOPS-2 User Registration DEVOPS-3 User Login DEVOPS-4 Password Reset	18	18

Conclusion: Successfully implemented the Agile Software Development Lifecycle using Jira by creating and managing a Scrum project. They learned to organize work using Epics, User Stories, Product Backlog, and Sprints, assign tasks to team members, monitor project progress through the Scrum Board, and analyze project performance using Jira reports. This experiment provided practical exposure to Agile project management and collaborative software development.